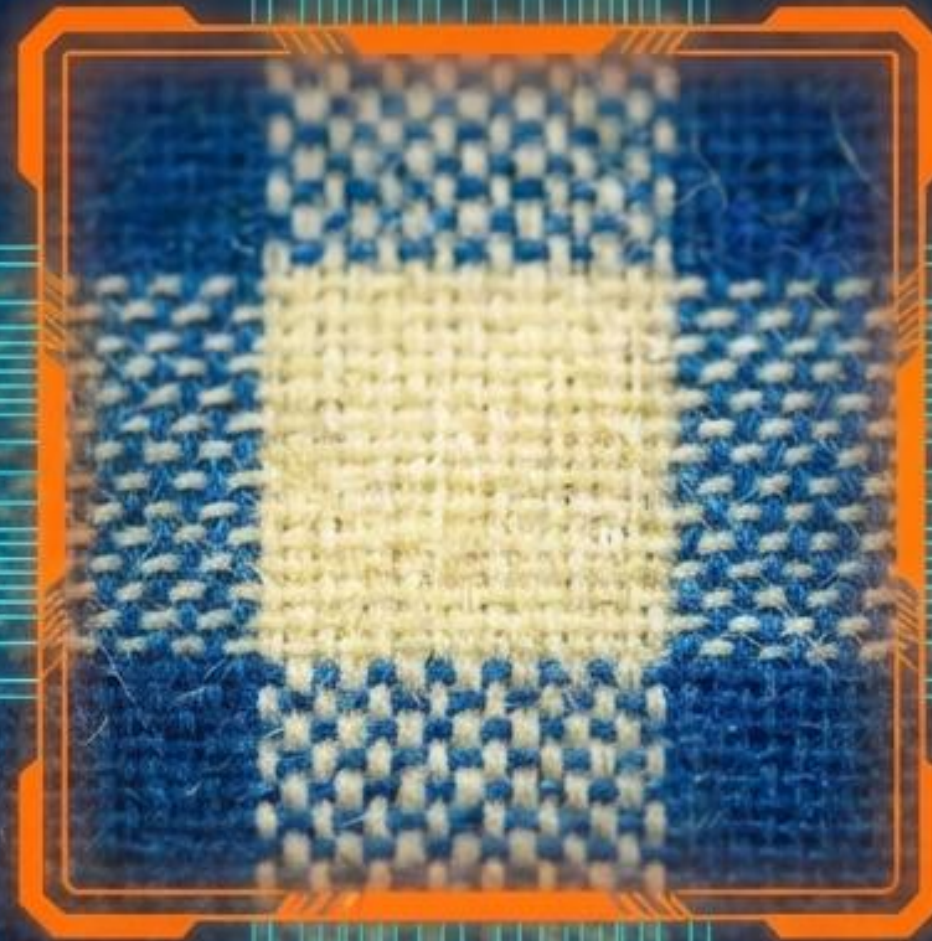


CLOSING THE LOOP: FROM DETECTION TO ACTION

The **Checkered Fabric AOI Digital Twin & Quality Assistant**

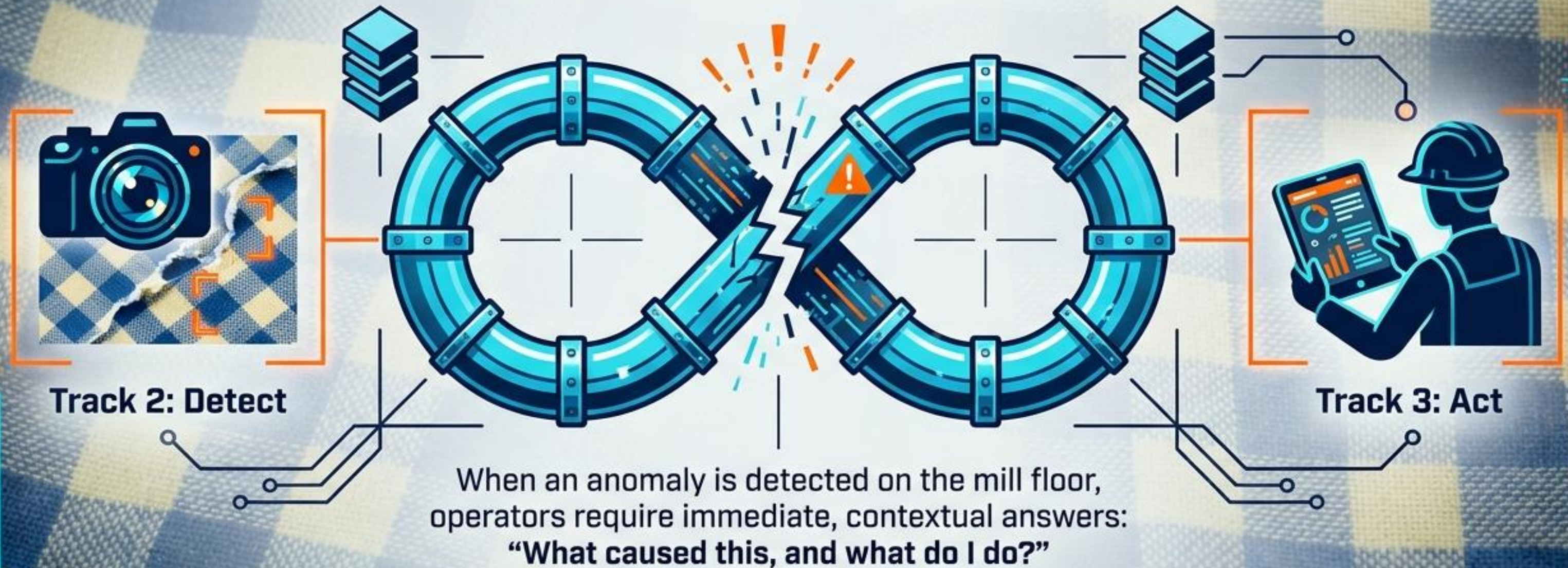


SMART FACTORIES

A.Y. 2026-2027 | UNIVERSITÀ DEGLI STUDI
DI ROMA TOR VERGATA

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Detection without action creates a manufacturing bottleneck



Our dual-track solution unites a **ResNet18 Defect Classifier** with a **RAG-based AI Quality Assistant** to seamlessly close the manufacturing loop.

Closing the loop requires a structural paradigm shift

Traditional AOI (The Open Loop)

Output

Provides binary pass/fail without context.

Operator Burden

High. Operators must manually search paper manuals or terminal PDFs to find corrective actions.

Knowledge Integration

Siloed data. The camera does not talk to the quality manual.

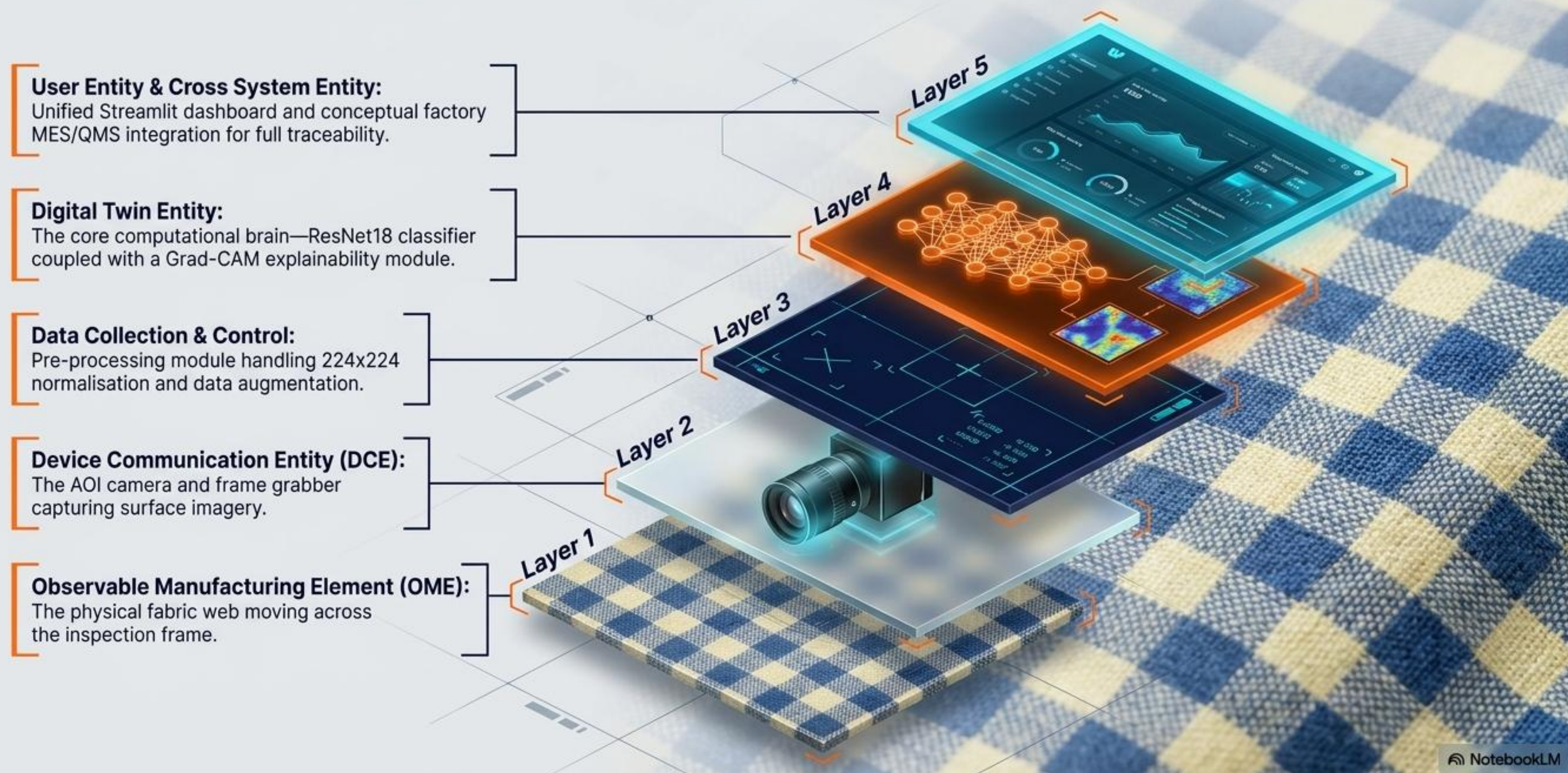
This Digital Twin (The Closed Loop)

Provides **exact defect class** backed by **visual explainability**.

Zero. Auto-surfaces the exact relevant SOP instantly.

Fully mapped to **ISO 23247 architecture**.

Architecting a true digital twin via ISO 23247



The observable element demands precise geometric learning

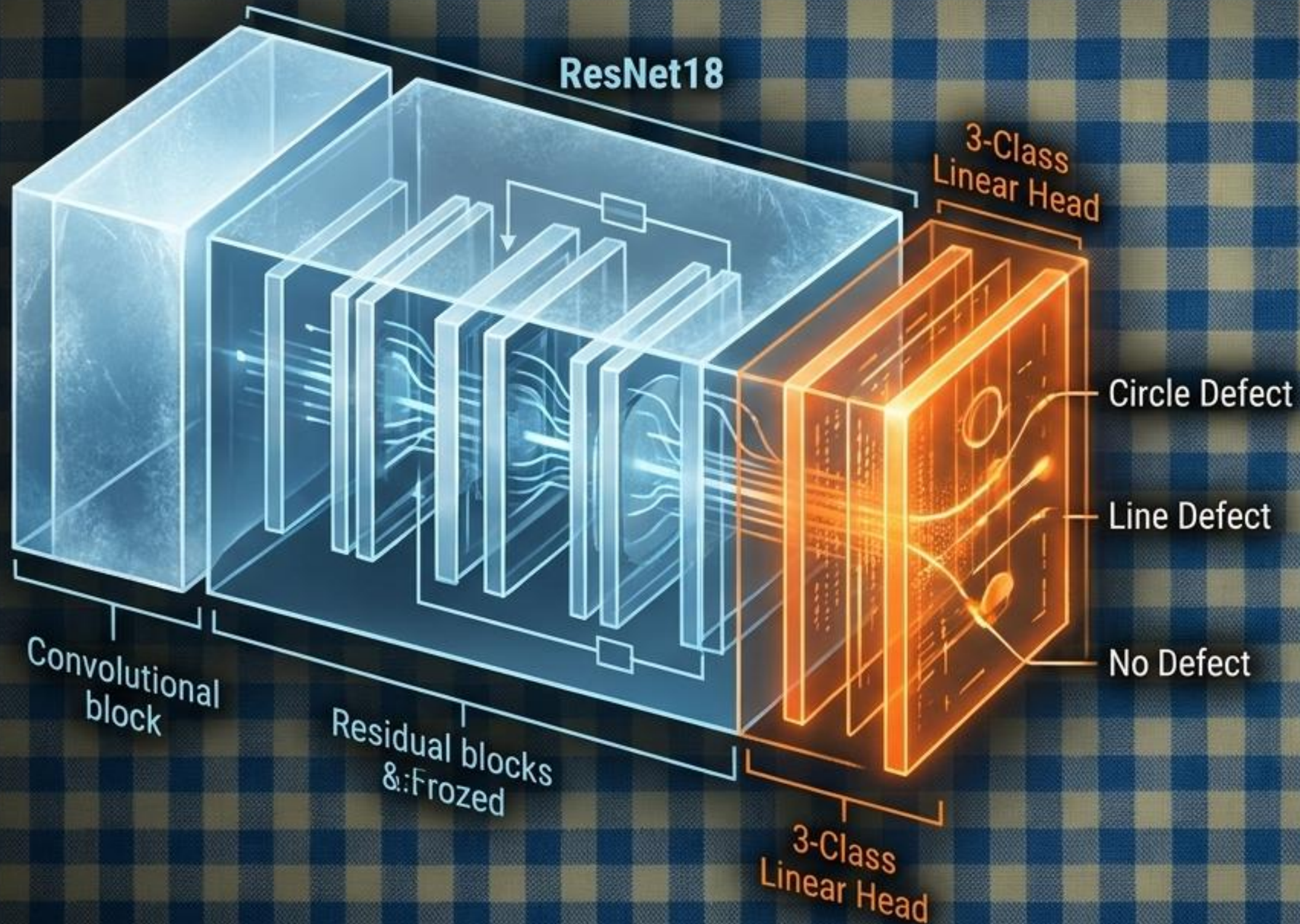


- 150 perfectly balanced photographs (50 per class) captured at **3072 x 3072 px**.
- Flaws were simulated using **physical objects** on healthy fabric to create **distinct defect signatures**.

Insight Callout

Exploratory Data Analysis confirms no class exhibits **systematically different mean brightness**. The model is forced to **learn the actual geometry** of the faults, preventing reliance on brightness shortcuts.

Transfer learning solves the small-dataset constraint



Methodology & Rationale



Frozen Backbone: ImageNet features (edges, textures, shapes) transfer exceptionally well to fabric patterns.

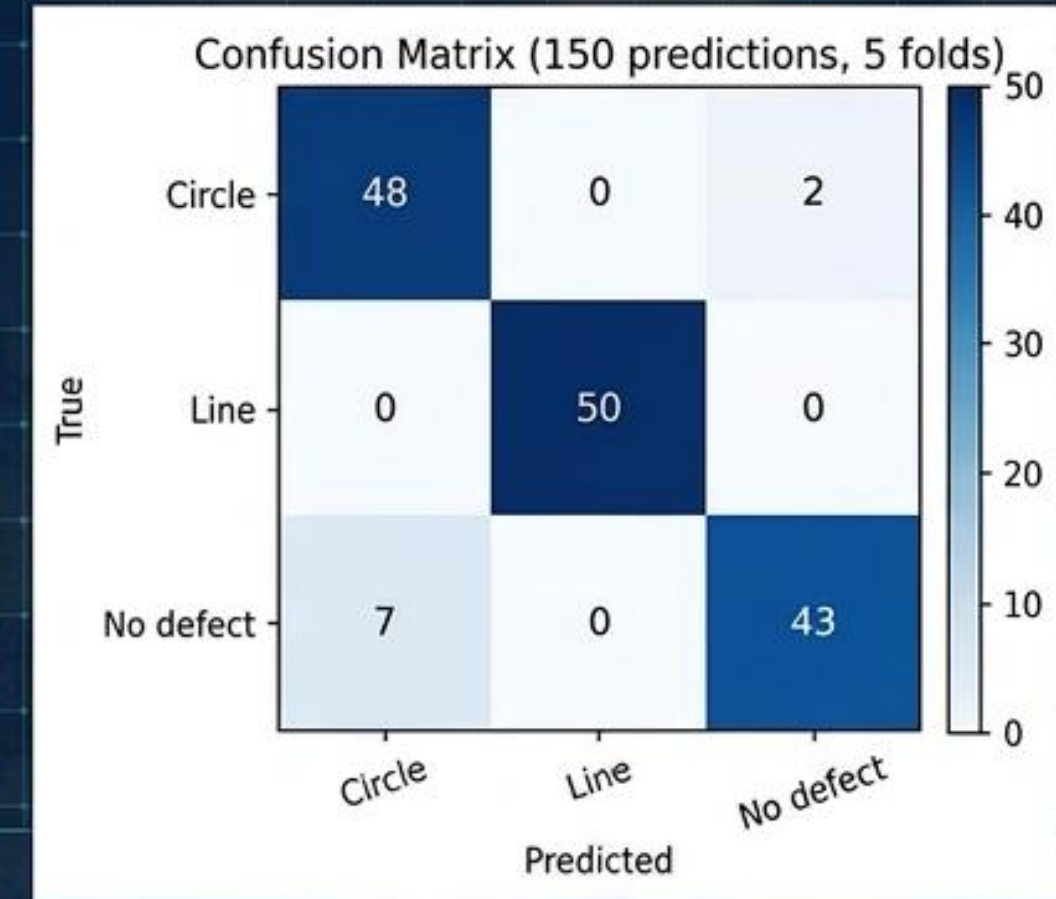
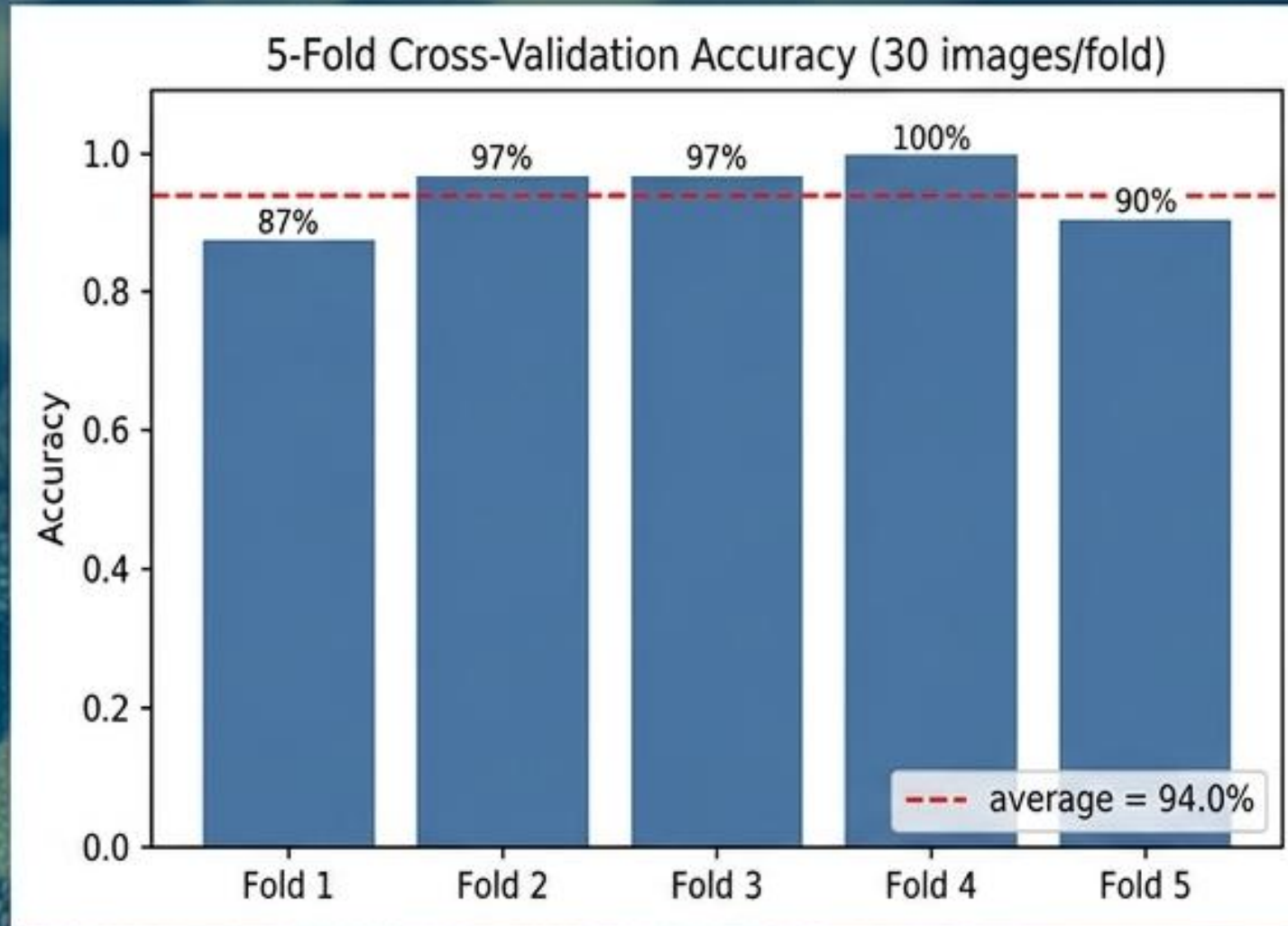


Active Head: Only the final 3-class linear head is trained. Solves the small-dataset problem, preventing severe overfitting.



Augmentation Strategy: Random horizontal flip, rotation ($\pm 10^\circ$), and colour jitter applied during training to ensure robustness.

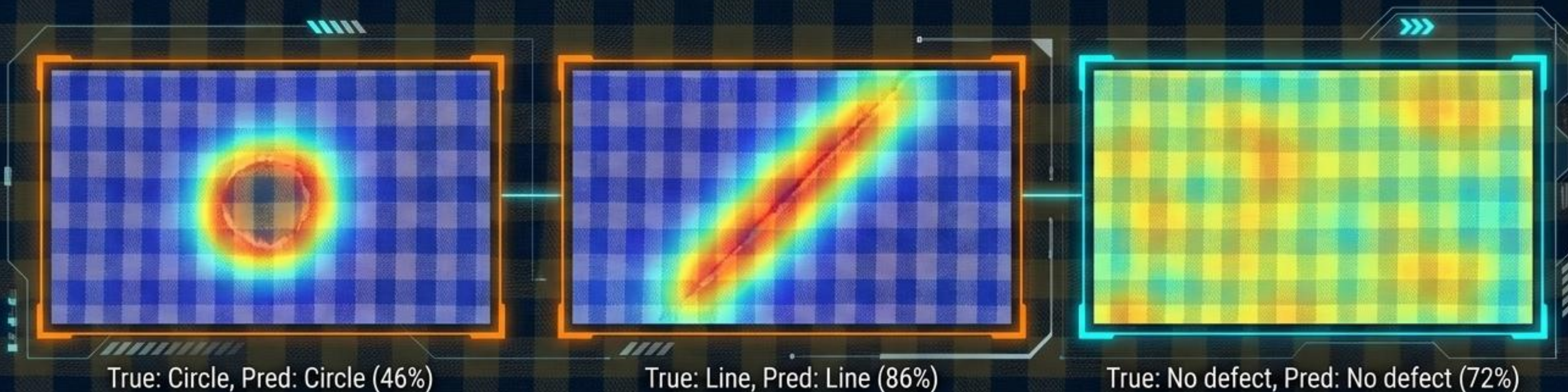
Robust validation yields 94% average classification accuracy



INSIGHT

1. **94.0% Average Accuracy** ($\pm 4.9\%$ standard deviation) via Stratified 5-Fold Cross-Validation.
2. Perfect Recall: 1.00 precision and recall for "Line" defects. Linear scratches are highly distinctive against the weave.
3. The 87%–100% per-fold range is expected sampling noise. Minor confusion primarily exists between faint "Circle" holes and natural "No defect" weave gaps.

Visual proof builds essential operator trust



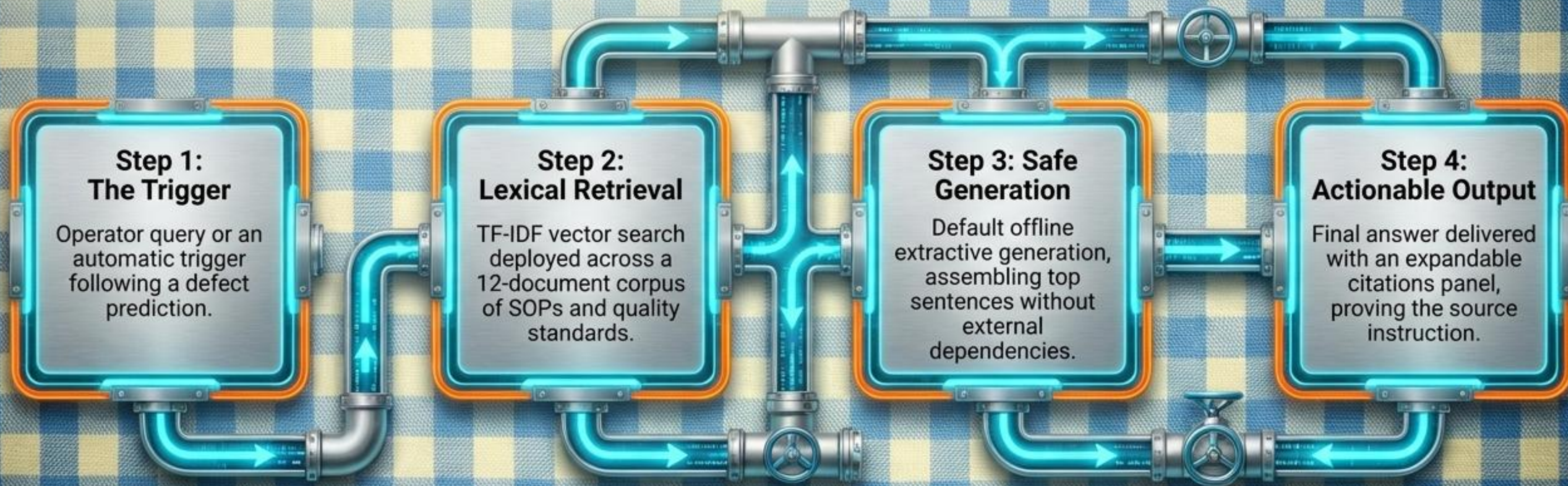
The Mechanism

Grad-CAM applied to the last residual block (layer 4). We don't just predict; we physically map the decision boundary.

Operator Value

The heatmap highlights the exact pixel regions driving the predicted class. This guarantees the operator can instantly verify the AI is reacting to physical fabric defects, rather than anomalous lighting, staging shadows, or background artefacts.

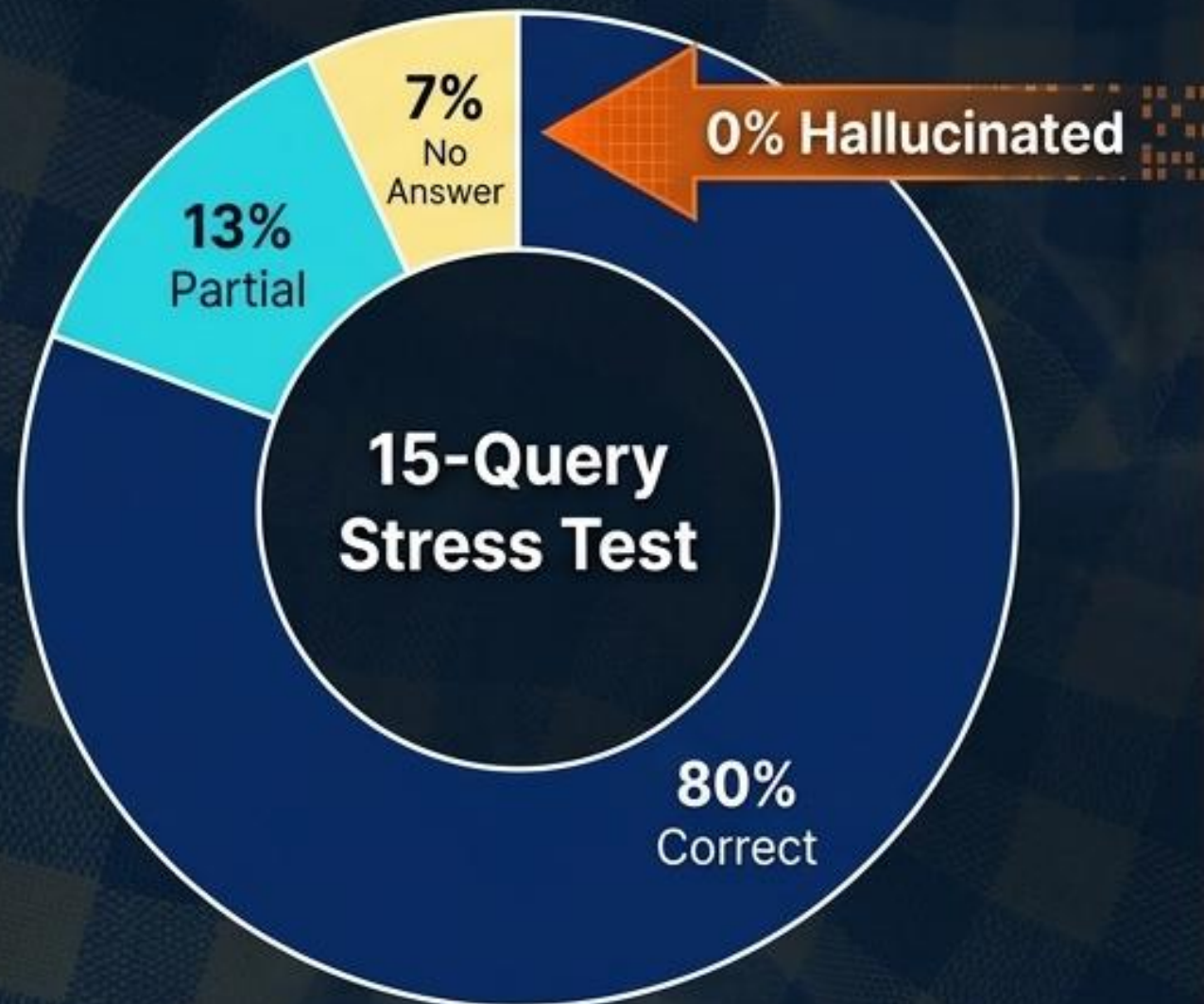
Natural-language retrieval digitises industrial knowledge



Key Industrial Constraint Met: The default architecture is fully offline. Zero data leaves the factory floor, and no API costs are incurred.

Offline extractive generation eliminates hallucination risk

15-Query Stress Test



Flawless Execution (12/15):

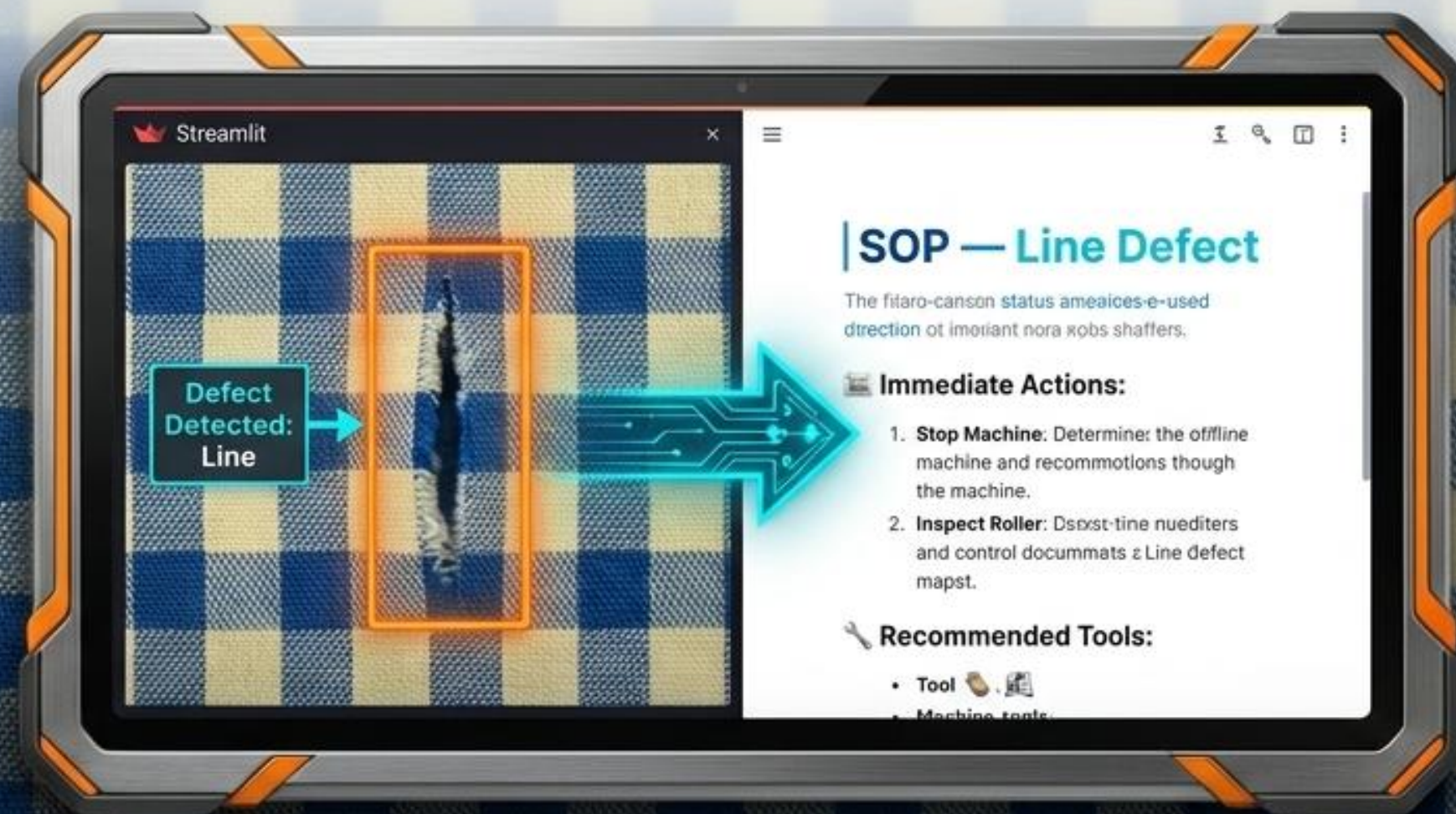
Successfully retrieves correct SOPs, maintenance guides, and ISO mappings for single-document queries.

Safe Rejection (1/15): For out-of-scope queries (e.g., 'market price of cotton yarn'), the system safely rejects and returns "No Answer" rather than fabricating data.

The Safety Guarantee (0/15):

Because the extractive composer strictly returns real corpus sentences, hallucination risk is structurally eliminated.

A unified dashboard connects anomaly detection directly to corrective action



See:

The camera captures a tear on the finishing frame.



Analyze:

The ResNet18 model classifies it as a "Line Defect" and maps it.



Bridge:

The prediction automatically queries the offline knowledge base.



Act:

The RAG Assistant instantly surfaces immediate corrective instructions.

Takeaway: The system transitions the operator's role from **diagnosing** a problem to **executing a solution**.

Prototype realities demand critical evaluation before scale

LIMITATIONS

SCALE & FUTURE NEEDS

Material Uniformity

The model has only seen one blue/yellow check pattern. **New weave geometries or colours mandate additional labelled data and retraining.**



Staged Defects vs. Reality

Circle/Line faults were simulated using physical objects. Real weaving pulls or machine abrasions may present **different edge signatures.**



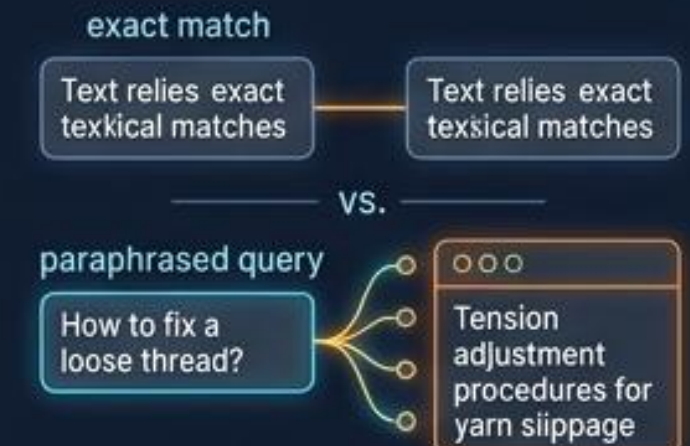
Corpus Scale

The 12-document knowledge base is illustrative. Real factory deployment requires ingesting sprawling, highly complex mill manuals.



Retrieval Limits

TF-IDF relies on exact lexical matches and struggles with highly paraphrased operator queries compared to semantic embedding models.



Deploying to the mill floor requires structural resilience

1. Confidence Routing (QA Safety Net)

Implement a strict 0.7–0.8 confidence threshold. Any prediction falling below this margin bypasses automation and is immediately routed to a human quality engineer.



2. Cross System Integration (Traceability)

Map the Digital Twin to live operations by logging every prediction (roll ID, timestamp, class, Grad-CAM proof) directly to the factory MES/QMS for Statistical Process Control.



3. Network Independence (Uptime)

Maintain the default TF-IDF offline architecture to ensure the AOI station and Quality Assistant remain 100% functional during inevitable factory network outages.



3100 60AS 31A2 3A220

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A completely integrated standard for automated optical inspection

Visually Accurate

Achieves 94% defect classification, backed by pixel-level Grad-CAM proof to secure operator trust.

Instantly Actionable

Replaces manual lookup with a zero-hallucination RAG assistant that serves precise corrective SOPs precisely when needed.

Architecturally Sound

Built explicitly upon the ISO 23247 framework, establishing a true Digital Twin rather than an isolated algorithm.

This dual-track architecture bridges the gap between seeing a problem on the loom and solving it on the floor.

(Live demo available via `streamlit run app.py`)